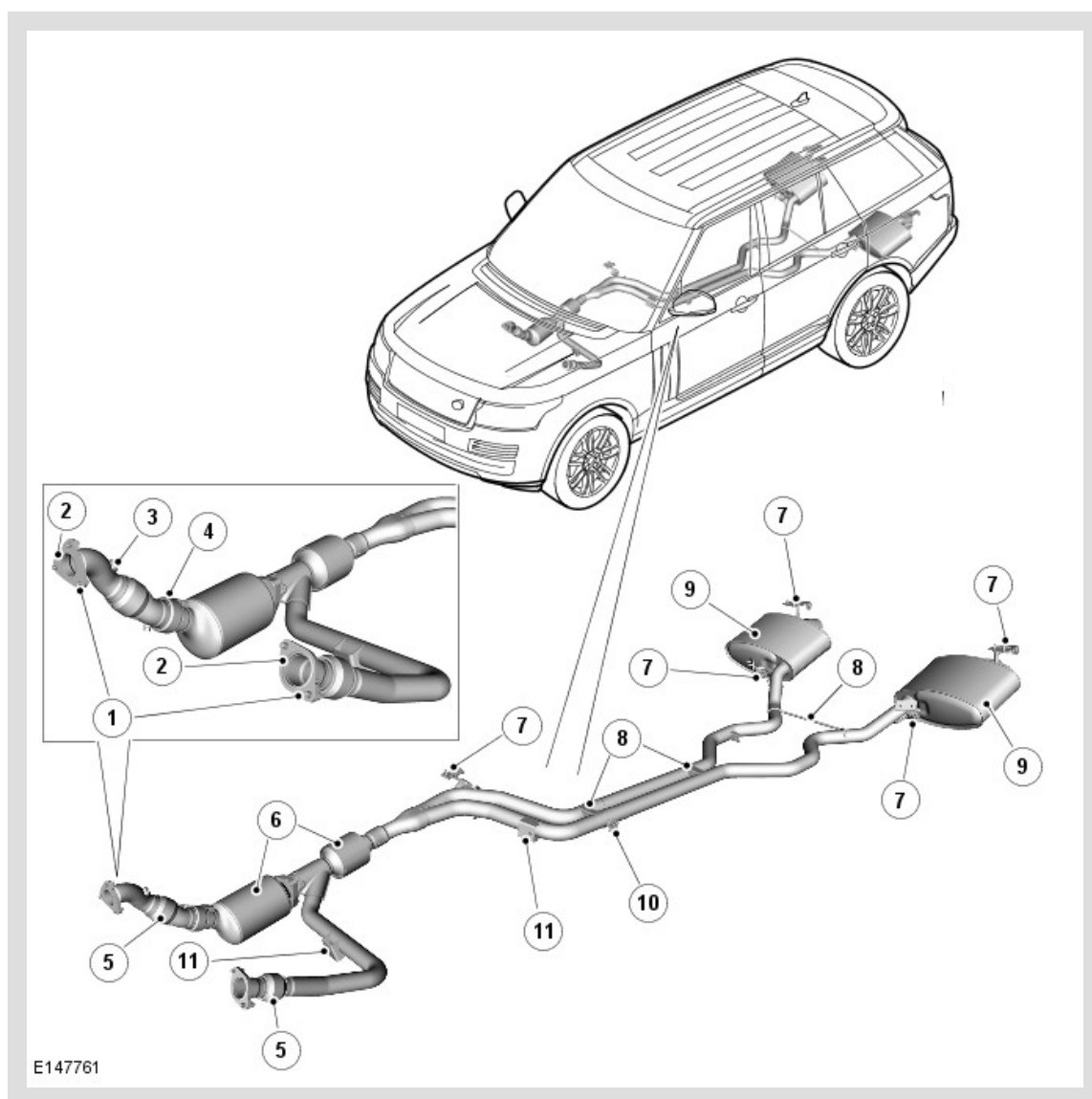


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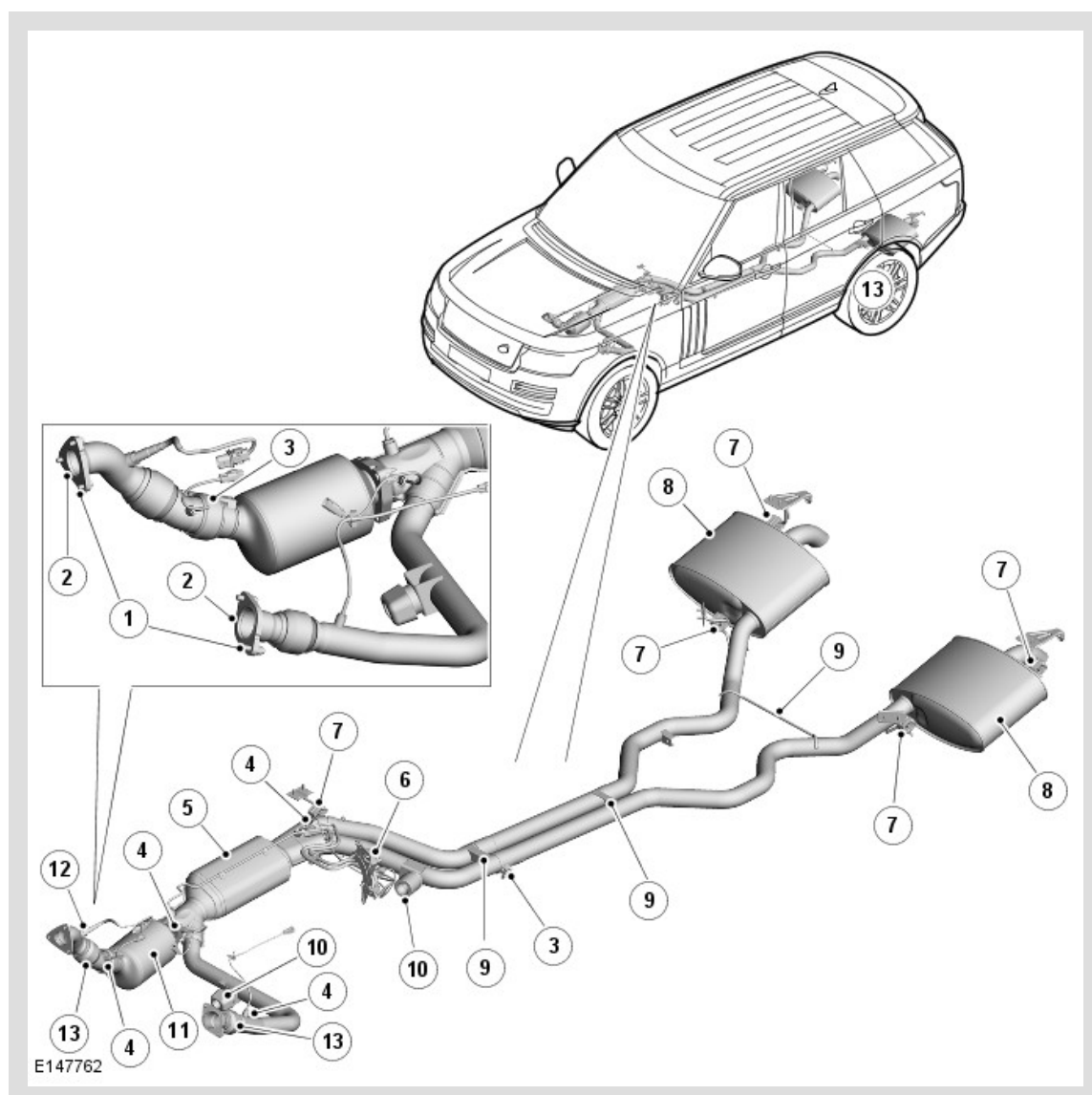
EXHAUST SYSTEM - TDV8 4.4L DIESEL

TDV8 4.4L DIESEL - EXHAUST SYSTEM COMPONENT LOCATION -
WITHOUT DIESEL PARTICULATE FILTER

ITEM	DESCRIPTION
1	Bolt (5 Off)
2	Gasket
3	Heated Oxygen Sensor Mounting Boss
4	Clamp
5	Flexible Coupling
6	Catalytic Converter

7	Mounting Rubber (5 Off)
8	Exhaust Brace
9	Silencer - Rear
10	Torca Clamp
11	Mass Damper (2 Off)

TDV8 4.4L DIESEL - EXHAUST SYSTEM COMPONENT LOCATION - WITH DIESEL PARTICULATE FILTER



ITEM	DESCRIPTION
1	Bolts (5 Off)
2	Gasket
3	Clamp
4	Exhaust Gas Temperature Sensor

5	Diesel Particulate Filter
6	Differential Pressure Sensor
7	Mounting Rubber (5 Off)
8	Silencer - Rear
9	Exhaust Brace
10	Mass Damper
11	Catalytic Converter
12	Heated Oxygen Sensor Mounting Boss
13	Flexible Coupling

OVERVIEW

There are two variants of the TDV8 4.4L diesel exhaust system:

- Non DPF System – EU4 Emission regulations.
- DPF System – EU5 Emission regulations.

On vehicles with a DPF fitted, a single catalytic converter is located in the bank 1 exhaust downpipe from the manifold. On vehicles without a DPF, a second catalytic converter is fitted in place of the DPF.

The system is attached to the underside of the body with 5 mounting rubbers which are located on hanger brackets that are welded to the system. The mounting rubbers locate on hanger brackets which are bolted to the underside of the vehicle body.

⚠ CAUTIONS:

- The use of bio-fuels can seriously contaminate and destroy the coatings used on the catalytic converter. The DPF and the catalytic converter can become irreversibly contaminated if non-specified oils or fuels are used. This will result in the vehicle being unable to regenerate the DPF, becoming non-compliant with rear silencer emission regulations and replacement of the catalytic converter and DPF will be required.
- If the vehicle is waded in deep water and the engine is stopped with the rear silencers submerged, the water, which can enter the system, can also contaminate both the DPF and the catalytic converter. This can result in catalytic converter damage and damaging the ability for the DPF to regenerate therefore requiring both components to be replaced.

COMPONENT DESCRIPTION

FRONT SECTION

The front section comprises of two catalytic converters or if a DPF is fitted a catalytic converter and diesel particulate filter. The bank 1 exhaust inlet pipe connects to the catalytic converter and the bank 2 exhaust inlet pipe connects in between the two catalytic converters or the catalytic converter and diesel particulate filter (if fitted).

Each inlet pipe is fitted with a flange that connects to the turbocharger for each cylinder bank. The bank 1 inlet flange uses a 3 bolt fixing and the bank 2 inlet flange uses a 2 bolt fixing that are screwed into threaded holes in each of the turbochargers. Each flange is sealed using a gasket and both inlet pipes are fitted with flexible couplings.

The catalytic converter is joined to either a second catalytic converter or a diesel particulate filter using a 3 bolt fixing. The front section of the exhaust also has 2 mass dampers which absorbs resonance from the system.

Vehicles without DPF: The catalytic converter outlet pipe is connected to a 'Y' joint which splits the outlet pipe into 2 separate pipes which connect to the rear silencer.

Vehicles with DPF: The diesel particulate filter outlet pipe is connected to a 'Y' joint which splits outlet pipe into 2 separate pipes which connect to the rear silencer.

REAR SECTION

The rear section comprises of two pipes from the catalytic converter or diesel particulate filter that go into each of the rear silencers. They are joined at the centre point of the exhaust using a torca clamp.

The two rear silencers are attached to the body using two exhaust hanging brackets on both sides of the silencers.

There are 3 brace's that are welded between the pipes, to retain system in position and reduce exhaust flexing.

CATALYTIC CONVERTER

The oxidizing catalytic converter is located in the front section of bank 1 inlet pipe from the turbocharger, after the pre-catalyst exhaust temperature sensor (vehicles with DPF only) and the Heated Oxygen Sensor (HO2S).



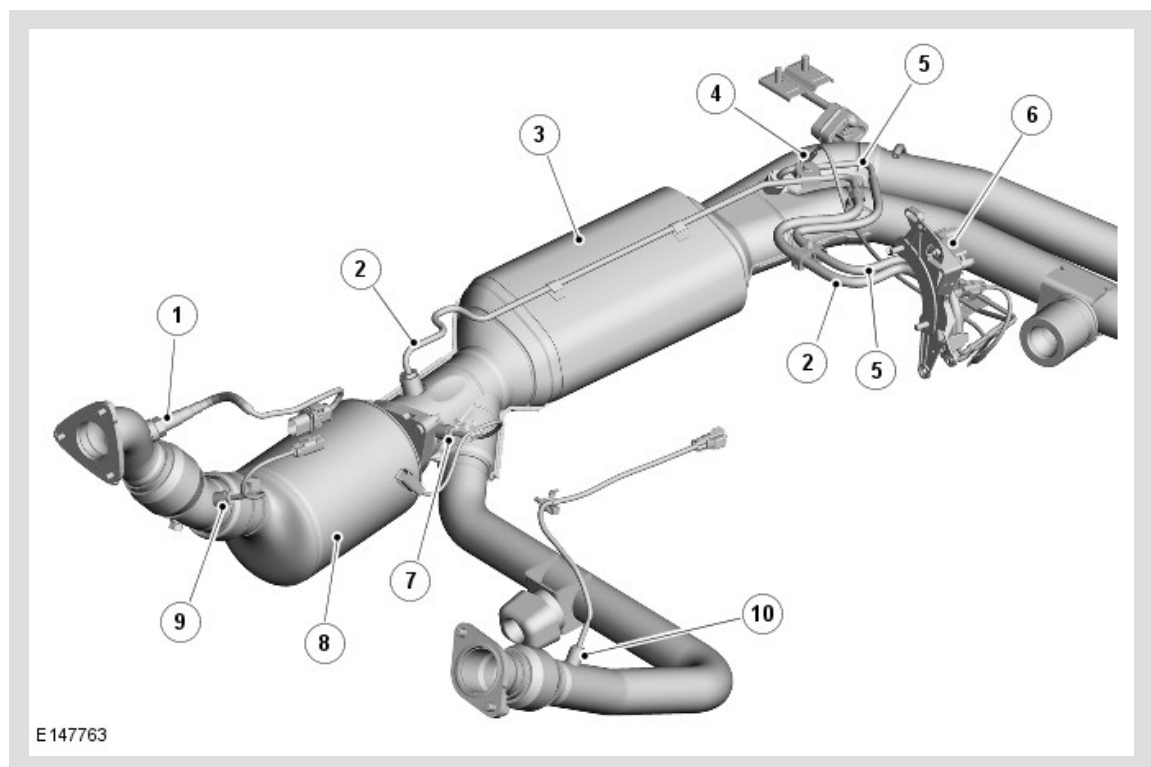
NOTE:

On vehicles fitted with DPF: The Pre and Post DPF Temperature Sensors are used by the engine management system to monitor the DPF regeneration purposes.

On vehicles without a DPF, a second catalytic converter is fitted in place of the DPF.

The engine management system provides accurately metered quantities of fuel to the combustion chambers to ensure the most efficient use of fuel and to minimise the exhaust emissions. On vehicles without DPF, a second catalytic converter is fitted to further reduce the carbon monoxide and the hydrocarbons content of the gases. In the catalytic converter the exhaust gases are passed through honeycombed ceramic elements coated with a special surface treatment called 'washcoat'. The washcoat increases the surface area of the ceramic elements by a factor of approximately 7000. The washcoat is a coating containing platinum and palladium, which are active constituents for converting harmful emissions into inert by-products. The platinum and palladium add oxygen to the carbon monoxide and the hydrocarbons in the exhaust gases and convert them into carbon dioxide and water respectively.

DIESEL PARTICULATE FILTER



ITEM	DESCRIPTION
1	Heated Oxygen Sensor
2	High Pressure Pipe
3	Diesel Particulate Filter
4	Post-DPF Temperature Sensor
5	Low Pressure Pipe
6	Differential Pressure Sensor
7	Post-Catalyst Temperature Sensor
8	Catalytic Converter
9	Pre-Catalyst Temperature Sensor
10	Pre-DPF Temperature Sensor

For further information regarding Diesel Particulate Filter refer to Exhaust System (309-00B: Exhaust System - TDV8 4.4L Diesel, Diesel Particulate Filter (System Operation and Component Description))